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EXPERIMENT 10

EXPLORE ANTIVIRUS DETECTION TECHNIQUES

AIM:

To understand the concept of antivirus software, its working principles, types of malware it detects, and its role in protecting computer systems from threats.

PROCEDURE:

1. Log in to the TryHackMe platform and start the Introduction to Antivirus room.
2. Study the basic concepts of malware and the need for antivirus software in system security.
3. Understand how antivirus software works, including signature-based and behavior-based detection methods.
4. Explore different types of malware such as viruses, worms, trojans, and ransomware.
5. Learn about antivirus features like real-time protection, scanning, quarantine, and updates.
6. Analyze how antivirus software detects, prevents, and removes threats from the system.
7. Attempt the tasks in the room and record observations on how antivirus tools enhance system security.

What is AV software?

Antivirus (AV) software is an extra layer of security that aims to detect and prevent the execution and spread of malicious files in a target operating system.

It is a host-based application that runs in real-time (in the background) to monitor and check the current and newly downloaded files. The AV software inspects and decides whether files are malicious using different techniques, which will be covered later in this room.

Interestingly, the first antivirus software was designed solely to detect and remove computer viruses.

Nowadays, that has changed; modern antivirus applications can detect and remove computer viruses as well as other harmful files and threats.

What does AV software look for?

Traditional AV software looks for **malware** with predefined malicious patterns or signatures. Malware is harmful software whose primary goal is to cause damage to a target machine, including but not limited to:

- Gain full access to a target machine.
- Steal sensitive information such as passwords.
- Encrypt files and cause damage to files.
- Inject other malicious software or unwanted advertisements.
- Used the compromised machine to perform further attacks such as botnet attacks.



AV vs other security products



In addition to AV software, other host-based security solutions provide real-time protection to endpoint devices. Endpoint Detection and Response (EDR) is a security solution that provides real-time protection based on behavioral analytics. An antivirus application performs scanning, detecting, and removing malicious files. On the other hand, EDR monitors various security checks in the target machine, including file activities, memory, network connections, Windows registry, processes, etc.

Modern Antivirus products are implemented to integrate the traditional Antivirus features and other advanced functionalities (similar to EDR functionalities) into one product to provide comprehensive protection against digital threats. For more information about Host-based security solutions, we suggest visiting the THM room: [The Lay of the Land](#).

AV software in the past and present

McAfee Associates, Inc. started the first AV software implementation in 1987. It was called "VirusScan," and its main goal at that time was to remove a virus named "Brain" that infected John McAfee's computer. Later, other companies joined in the battle against viruses. AV software was called scanners, and they were command-line software that searched for malicious patterns in files.

Since then, things have changed. AV software nowadays uses a Graphical User Interface (GUI) to perform scans for malicious files and other tasks. Malware programs have also expanded in scope and now target victims on Windows and other operating systems. Modern AV software supports most devices and platforms, including Windows, Linux, macOS, Android, and iOS. Modern AV software has improved and become more intelligent and sophisticated, as they pack a bundle of versatile features, including Antivirus, Anti-Exploit, Firewall, Encryption tool, etc.

We will be discussing some AV features in the next task.

Answer the questions below

What does AV mean?

✓ Correct Answer

Which PC Antivirus vendor implemented the first AV software on the market?

✓ Correct Answer

Antivirus software is a _____-based security solution.

✓ Correct Answer

Answer the questions below

Which AV feature analyzes malware in a safe and isolated environment?

✓ Correct Answer

An _____ feature is a process of restoring or decrypting the compressed executable files to the original.

✓ Correct Answer

Read the above to proceed to the next task, where we discuss the AV detection techniques.

✓ Correct Answer

Answer the questions below

What is the `sigtool` tool output to generate an MD5 of the `AV-Check.exe` binary?

✓ Correct Answer



Use the strings tool to list all human-readable strings of the AV-Check binary. What is the flag?

✓ Correct Answer



Answer the questions below

Which detection method is used to analyze malicious software inside virtual environments?

✓ Correct Answer

Task 8 Conclusion

Recap

In this room, we covered Antivirus software and its detection approaches. As a red teamer, it is important to know how AV software works and detects malicious applications to be able to implement bypass techniques.

We also discussed the static detection technique in detail and showcased how ClamAV, an open-source Antivirus, detects malicious files using static analysis. Additionally, we showed how to create your own database and use Yara rules to detect malicious files that are not detected by the official database.

Once we obtain access to a target, it is essential to enumerate the target machine before performing further actions, such as privilege escalation or lateral movement. The reason is to not trigger alerts for suspicious activities, which may cause losing access to the target machine. Thus, we introduced two methods to practice AV fingerprinting using public and private tools.

Answer the questions below

Congrats on completing the room, and keep learning!

✓ Correct Answer



Introduction to Antivirus

Understand how antivirus software works and what detection techniques are used to bypass malicious file checks.

📶 🔄 ⌚ 90 min 👤 31,417 🌐

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Room completed (100%)

Task 1 Introduction

Task 2 Antivirus Software

Task 3 Antivirus Features

Task 4 Deploy the VM

Task 5 AV Static Detection

Task 6 Other Detection Techniques

Task 7 AV Testing and Fingerprinting

Task 8 Conclusion

RESULT:

The antivirus detection software was explored and tasks were completed successfully.